



1. Description

1.1. Project

Project Name	STM32G473RET6
Board Name	custom
Generated with:	STM32CubeMX 6.15.0
Date	03/22/2026

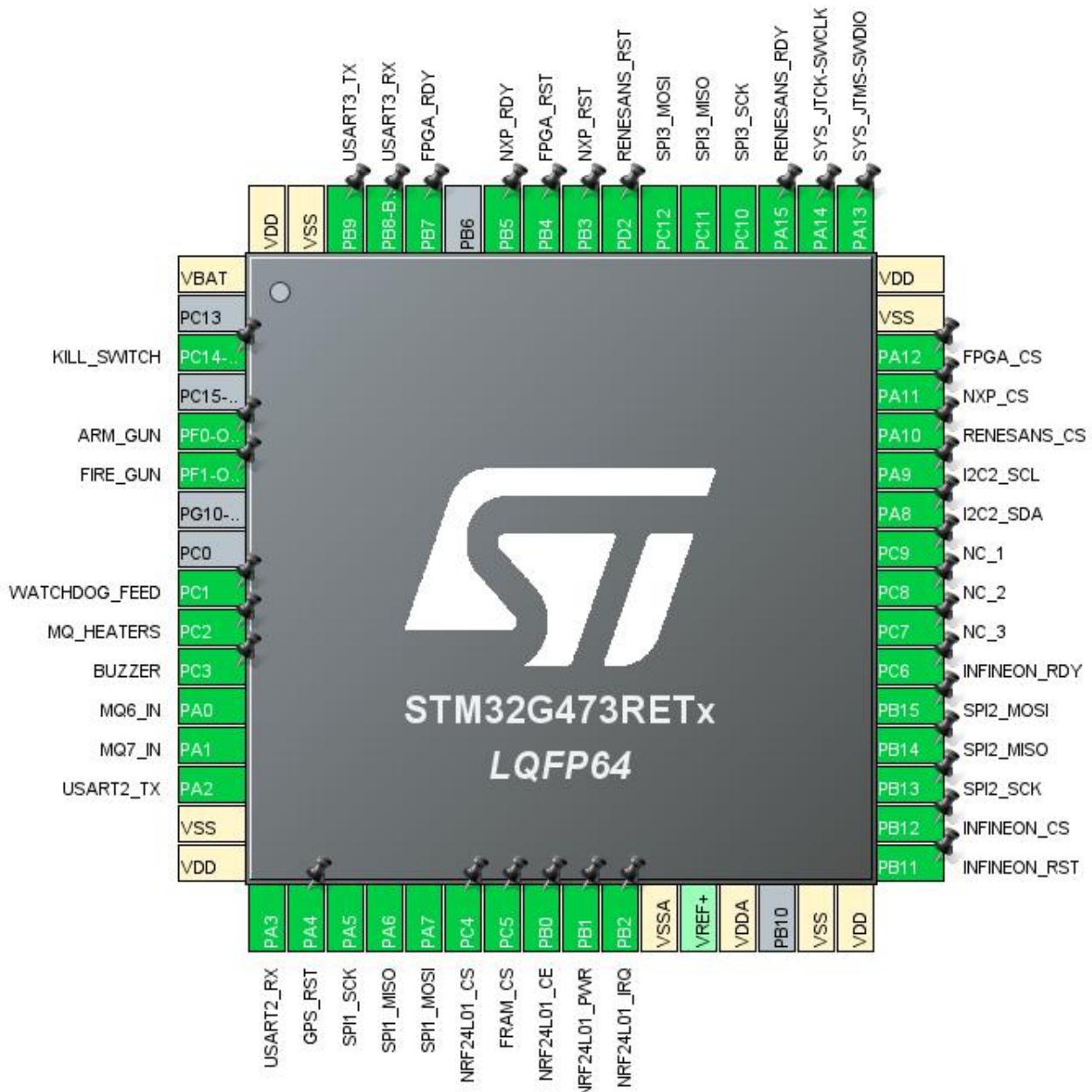
1.2. MCU

MCU Series	STM32G4
MCU Line	STM32G4x3
MCU name	STM32G473RETx
MCU Package	LQFP64
MCU Pin number	64

1.3. Core(s) information

Core(s)	ARM Cortex-M4
---------	---------------

2. Pinout Configuration



3. Pins Configuration

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
3	PC14-OSC32_IN *	I/O	GPIO_Output	KILL_SWITCH
5	PF0-OSC_IN *	I/O	GPIO_Output	ARM_GUN
6	PF1-OSC_OUT *	I/O	GPIO_Output	FIRE_GUN
9	PC1 *	I/O	GPIO_Output	WATCHDOG_FEED
10	PC2	I/O	TIM20_CH2	MQ_HEATERS
11	PC3	I/O	TIM1_CH4	BUZZER
12	PA0	I/O	ADC1_IN1	MQ6_IN
13	PA1	I/O	ADC1_IN2	MQ7_IN
14	PA2	I/O	USART2_TX	
15	VSS	Power		
16	VDD	Power		
17	PA3	I/O	USART2_RX	
18	PA4 *	I/O	GPIO_Output	GPS_RST
19	PA5	I/O	SPI1_SCK	
20	PA6	I/O	SPI1_MISO	
21	PA7	I/O	SPI1_MOSI	
22	PC4 *	I/O	GPIO_Output	NRF24L01_CS
23	PC5 *	I/O	GPIO_Output	FRAM_CS
24	PB0 *	I/O	GPIO_Output	NRF24L01_CE
25	PB1 *	I/O	GPIO_Output	NRF24L01_PWR
26	PB2	I/O	GPIO_EXTI2	NRF24L01_IRQ
27	VSSA	Power		
29	VDDA	Power		
31	VSS	Power		
32	VDD	Power		
33	PB11 *	I/O	GPIO_Output	INFINEON_RST
34	PB12 *	I/O	GPIO_Output	INFINEON_CS
35	PB13	I/O	SPI2_SCK	
36	PB14	I/O	SPI2_MISO	
37	PB15	I/O	SPI2_MOSI	
38	PC6	I/O	GPIO_EXTI6	INFINEON_RDY
39	PC7 *	I/O	GPIO_Output	NC_3
40	PC8 *	I/O	GPIO_Output	NC_2
41	PC9 *	I/O	GPIO_Output	NC_1
42	PA8	I/O	I2C2_SDA	

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
43	PA9	I/O	I2C2_SCL	
44	PA10 *	I/O	GPIO_Output	RENESANS_CS
45	PA11 *	I/O	GPIO_Output	NXP_CS
46	PA12 *	I/O	GPIO_Output	FPGA_CS
47	VSS	Power		
48	VDD	Power		
49	PA13	I/O	SYS_JTMS-SWDIO	
50	PA14	I/O	SYS_JTCK-SWCLK	
51	PA15	I/O	GPIO_EXTI15	RENESANS_RDY
52	PC10	I/O	SPI3_SCK	
53	PC11	I/O	SPI3_MISO	
54	PC12	I/O	SPI3_MOSI	
55	PD2 *	I/O	GPIO_Output	RENESANS_RST
56	PB3 *	I/O	GPIO_Output	NXP_RST
57	PB4 *	I/O	GPIO_Output	FPGA_RST
58	PB5	I/O	GPIO_EXTI5	NXP_RDY
60	PB7	I/O	GPIO_EXTI7	FPGA_RDY
61	PB8-BOOT0	I/O	USART3_RX	
62	PB9	I/O	USART3_TX	
63	VSS	Power		
64	VDD	Power		

* The pin is affected with an I/O function



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32G4
Line	STM32G4x3
MCU	STM32G473RETx
Datasheet	DS12712 Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

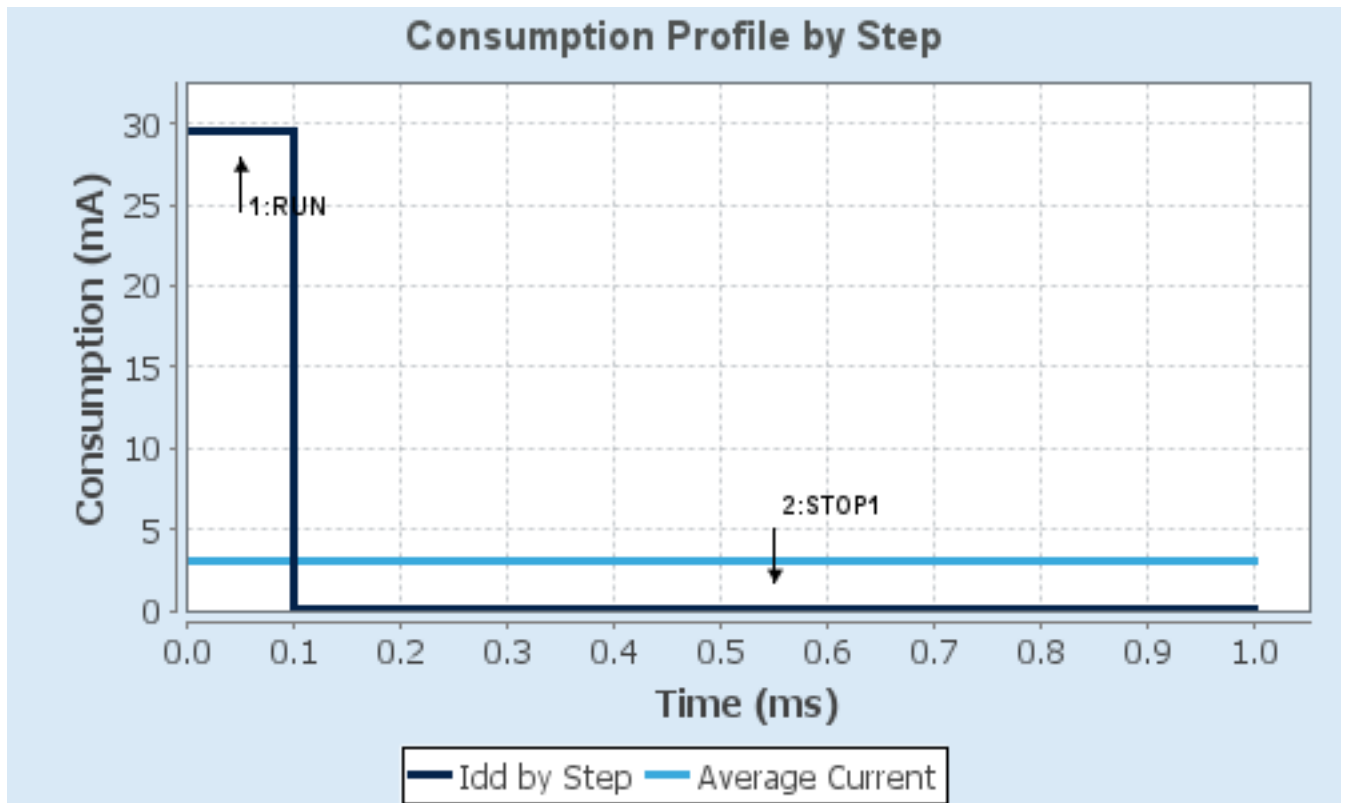
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP1
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-Boost	NoRange
Fetch Type	FLASH/DualBank/ART	NA
CPU Frequency	170 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL CLOCKS OFF
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	29.5 mA	80.5 μ A
Duration	0.1 ms	0.9 ms
DMIPS	213.0	0.0
Ta Max	124.25	129.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	3.02 mA
Battery Life	1 month, 16 days, 9 hours	Average DMIPS	212.5 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	STM32G473RET6
Project Folder	D:\Projekty\TrinityTrack6000\MainControlBoard\STM32\STM32G473RET6
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_G4 V1.6.2
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_CORDIC_Init	CORDIC
5	MX_SPI1_Init	SPI1
6	MX_SPI2_Init	SPI2
7	MX_I2C2_Init	I2C2
8	MX_USART2_UART_Init	USART2
9	MX_TIM3_Init	TIM3
10	MX_ADC1_Init	ADC1
11	MX_SPI3_Init	SPI3

Rank	Function Name	Peripheral Instance Name
12	MX_USART3_UART_Init	USART3
13	MX_TIM1_Init	TIM1
14	MX_TIM20_Init	TIM20

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN1: IN1 Single-ended

IN2: IN2 Single-ended

mode: Temperature Sensor Channel

mode: Vrefint Channel

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler Synchronous clock mode divided by 4

Resolution ADC 12-bit resolution

Data Alignment Right alignment

Gain Compensation 0

Scan Conversion Mode Enabled

End Of Conversion Selection End of single conversion

Low Power Auto Wait Disabled

Continuous Conversion Mode Disabled

Discontinuous Conversion Mode Disabled

DMA Continuous Requests Disabled

Overrun behaviour Overrun data preserved

ADC_Regular_ConversionMode:

Enable Regular Conversions Enable

Enable Regular Oversampling Disable

Number Of Conversion **4 ***

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Rank 1

Channel Channel 1

Sampling Time **92.5 Cycles ***

Offset Number No offset

Rank **2 ***

Channel **Channel 2 ***

Sampling Time **92.5 Cycles ***

Offset Number No offset

Rank **3 ***

Channel **Channel Vrefint ***

Sampling Time **92.5 Cycles ***

Offset Number	No offset
<u>Rank</u>	4 *
Channel	Channel Temperature Sensor *
Sampling Time	92.5 Cycles *
Offset Number	No offset
ADC_Injected_ConversionMode:	
Enable Injected Conversions	Disable
Analog Watchdog 1:	
Enable Analog WatchDog1 Mode	false
Analog Watchdog 2:	
Enable Analog WatchDog2 Mode	false
Analog Watchdog 3:	
Enable Analog WatchDog3 Mode	false

3.2. CORDIC mode: Activated

3.3. I2C2 I2C: I2C

3.3.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Fast Mode *
I2C Speed Frequency (KHz)	400
Rise Time (ns)	100
Fall Time (ns)	100
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x40621236 *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.4. RCC

3.4.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Disabled
Data Cache	Enabled
Flash Latency(WS)	4 WS (5 CPU cycle)

RCC Parameters:

HSI Calibration Value	64
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1 boost
-------------------------------	---------------------------------------

Peripherals Clock Configuration:

Generate the peripherals clock configuration	TRUE
--	------

3.5. SPI1

Mode: Full-Duplex Master

3.5.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits *
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	32 *
Baud Rate	5.3125 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software

3.6. SPI2

Mode: Full-Duplex Master

3.6.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits *
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	16 *
Baud Rate	10.625 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software

3.7. SPI3

Mode: Full-Duplex Master

3.7.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits *
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	16 *
Baud Rate	10.625 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software

3.8. SYS

Debug: Serial Wire

Timebase Source: TIM6

mode: save power of non-active UCPD - deactive Dead Battery pull-up

3.9. TIM1

Clock Source : Internal Clock

Channel4: PWM Generation CH4

3.9.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	169 *
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	999 *
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- COMP3	Disable
- COMP4	Disable
- COMP5	Disable
- COMP6	Disable
- COMP7	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
------------	---------

BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- COMP3	Disable
- COMP4	Disable
- COMP5	Disable
- COMP6	Disable
- COMP7	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable
--------------------	---------

PWM Generation Channel 4:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.10. TIM3

Clock Source : Internal Clock

3.10.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	169 *
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	999 *
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

3.11. TIM20

Clock Source : Internal Clock

Channel2: PWM Generation CH2

3.11.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	169 *
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	999 *
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High
BRK Filter (4 bits value)	0
BRK Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable
- COMP3	Disable
- COMP4	Disable
- COMP5	Disable
- COMP6	Disable
- COMP7	Disable

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High
BRK2 Filter (4 bits value)	0
BRK2 Sources Configuration	
- Digital Input	Disable
- COMP1	Disable
- COMP2	Disable

- COMP3	Disable
- COMP4	Disable
- COMP5	Disable
- COMP6	Disable
- COMP7	Disable

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

Clear Input:

Clear Input Source	Disable
--------------------	---------

PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.12. USART2

Mode: Asynchronous

3.12.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable

RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.13. USART3

Mode: Asynchronous

3.13.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.14. STMicroelectronics.X-CUBE-AZRTOS-G4.2.0.0

mode: RTOSJjThreadX

3.14.1. AzureRTOS Application:

Memory Configuration:

Memory Allocation	Use Static Allocation
ThreadX memory pool size	1024

ThreadX Core Init:

ThreadX Generate Init Code	true *
Create ThreadX Application Thread	false
Create ThreadX Application Semaphore	false
Create ThreadX Application Mutex	false
Create ThreadX Application MessageQueue	false

3.14.2. ThreadX:

Version:

ThreadX version	6.2.0
-----------------	-------

Core:

TX_MINIMUM_STACK	200
TX_THREAD_USER_EXTENSION	
TX_DISABLE_STACK_FILLING	Disabled
TX_ENABLE_STACK_CHECKING	Disabled
TX_DISABLE_PREEMPTION_THRESHOLD	Enabled
TX_DISABLE_REDUNDANT_CLEARING	Disabled
TX_DISABLE_NOTIFY_CALLBACKS	Enabled
TX_INLINE_THREAD_RESUME_SUSPEND	Disabled
TX_NOT_INTERRUPTABLE	Disabled
TX_MAX_PRIORITIES	32
TX_TIMER_TICKS_PER_SECOND	2000 *
ALIGN_TYPE_DEFINED	Disabled
TX_MEMSET	memset
TX_ENABLE_IAR_LIBRARY_SUPPORT	Disabled
TX_SAFETY_CRITICAL	Disabled
TX_NO_FILEX_POINTER	Disabled
Enable BASEPRI support	Disabled
TX_DISABLE_ERROR_CHECKING	Disabled

Timer:

TX_TIMER_PROCESS_IN_ISR	Disabled
TX_REACTIVATE_INLINE	Disabled
TX_TIMER_THREAD_STACK_SIZE	1024
TX_TIMER_THREAD_PRIORITY	0

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PA0	ADC1_IN1	Analog mode	No pull-up and no pull-down	n/a	MQ6_IN
	PA1	ADC1_IN2	Analog mode	No pull-up and no pull-down	n/a	MQ7_IN
I2C2	PA8	I2C2_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PA9	I2C2_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
SPI1	PA5	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	High *	
	PA6	SPI1_MISO	Alternate Function Push Pull	No pull-up and no pull-down	High *	
	PA7	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	High *	
SPI2	PB13	SPI2_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PB14	SPI2_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PB15	SPI2_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
SPI3	PC10	SPI3_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC11	SPI3_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC12	SPI3_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	
TIM1	PC3	TIM1_CH4	Alternate Function Push Pull	No pull-up and no pull-down	Low	BUZZER
TIM20	PC2	TIM20_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	MQ_HEATERS
USART2	PA2	USART2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA3	USART2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART3	PB8-BOOT0	USART3_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB9	USART3_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
GPIO	PC14-OSC32_IN	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	KILL_SWITCH
	PF0-OSC_IN	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ARM_GUN
	PF1-OSC_OUT	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	FIRE_GUN
	PC1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	WATCHDOG_FEED
	PA4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPS_RST
	PC4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	NRF24L01_CS

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PC5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	FRAM_CS
	PB0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	NRF24L01_CE
	PB1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	NRF24L01_PWR
	PB2	GPIO_EXTI2	External Interrupt Mode with Falling edge trigger detection	No pull-up and no pull-down	n/a	NRF24L01_IRQ
	PB11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	INFINEON_RST
	PB12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	INFINEON_CS
	PC6	GPIO_EXTI6	External Interrupt Mode with Falling edge trigger detection	No pull-up and no pull-down	n/a	INFINEON_RDY
	PC7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	NC_3
	PC8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	NC_2
	PC9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	NC_1
	PA10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RENESANS_CS
	PA11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	NXP_CS
	PA12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	FPGA_CS
	PA15	GPIO_EXTI15	External Interrupt Mode with Falling edge trigger detection	No pull-up and no pull-down	n/a	RENESANS_RDY
	PD2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	RENESANS_RST
	PB3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	NXP_RST
	PB4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	FPGA_RST
	PB5	GPIO_EXTI5	External Interrupt Mode with Falling edge trigger detection	No pull-up and no pull-down	n/a	NXP_RDY
	PB7	GPIO_EXTI7	External Interrupt Mode with Falling edge trigger detection	No pull-up and no pull-down	n/a	FPGA_RDY

4.2. DMA configuration

DMA request	Stream	Direction	Priority
SPI1_RX	DMA1_Channel4	Peripheral To Memory	High *
SPI1_TX	DMA1_Channel3	Memory To Peripheral	High *
SPI2_RX	DMA1_Channel5	Peripheral To Memory	Very High *
SPI2_TX	DMA1_Channel6	Memory To Peripheral	Very High *
I2C2_TX	DMA1_Channel1	Memory To Peripheral	Medium *
I2C2_RX	DMA1_Channel2	Peripheral To Memory	Medium *
ADC1	DMA2_Channel3	Peripheral To Memory	Low
USART2_TX	DMA1_Channel8	Memory To Peripheral	Low
USART2_RX	DMA1_Channel7	Peripheral To Memory	Low

SPI1_RX: DMA1_Channel4 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

SPI1_TX: DMA1_Channel3 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

SPI2_RX: DMA1_Channel5 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

SPI2_TX: DMA1_Channel6 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

I2C2_TX: DMA1_Channel1 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

I2C2_RX: DMA1_Channel2 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

ADC1: DMA2_Channel3 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: **Word ***
Memory Data Width: **Word ***

USART2_TX: DMA1_Channel8 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

USART2_RX: DMA1_Channel7 DMA request Settings:

Mode:	Circular *
Peripheral Increment:	Disable
Memory Increment:	Enable *
Peripheral Data Width:	Byte
Memory Data Width:	Byte

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	0	0
EXTI line2 interrupt	true	5	0
DMA1 channel1 global interrupt	true	8	0
DMA1 channel2 global interrupt	true	8	0
DMA1 channel3 global interrupt	true	2	0
DMA1 channel4 global interrupt	true	2	0
DMA1 channel5 global interrupt	true	5	0
DMA1 channel6 global interrupt	true	5	0
DMA1 channel7 global interrupt	true	0	0
EXTI line[9:5] interrupts	true	0	0
USART2 global interrupt / USART2 wake-up interrupt through EXTI line 26	true	0	0
EXTI line[15:10] interrupts	true	0	0
TIM6 global interrupt, DAC1 and DAC3 channel underrun error interrupts	true	15	0
DMA2 channel3 global interrupt	true	3	0
DMA1 channel8 global interrupt	true	0	0
PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/38/39/40/41	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
ADC1 and ADC2 global interrupt	unused		
TIM1 break interrupt and TIM15 global interrupt	unused		
TIM1 update interrupt and TIM16 global interrupt	unused		
TIM1 trigger and commutation interrupts and TIM17 global interrupt	unused		
TIM1 capture compare interrupt	unused		
TIM3 global interrupt	unused		
I2C2 event interrupt / I2C2 wake-up interrupt through EXTI line 24	unused		

Interrupt Table	Enable	Preenmption Priority	SubPriority
I2C2 error interrupt		unused	
SPI1 global interrupt		unused	
SPI2 global interrupt		unused	
USART3 global interrupt / USART3 wake-up interrupt through EXTI line 28		unused	
SPI3 global interrupt		unused	
TIM20 break interrupt		unused	
TIM20 update interrupt		unused	
TIM20 trigger and commutation interrupts		unused	
TIM20 capture compare interrupt		unused	
FPU global interrupt		unused	
CORDIC interrupt		unused	

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
EXTI line2 interrupt	false	true	true
DMA1 channel1 global interrupt	false	true	true
DMA1 channel2 global interrupt	false	true	true
DMA1 channel3 global interrupt	false	true	true
DMA1 channel4 global interrupt	false	true	true
DMA1 channel5 global interrupt	false	true	true
DMA1 channel6 global interrupt	false	true	true
DMA1 channel7 global interrupt	false	true	true
EXTI line[9:5] interrupts	false	true	true
USART2 global interrupt / USART2 wake-up interrupt through EXTI line 26	false	true	true
EXTI line[15:10] interrupts	false	true	true
TIM6 global interrupt, DAC1 and DAC3 channel underrun error interrupts	false	true	true
DMA2 channel3 global interrupt	false	true	true
DMA1 channel8 global interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

Middleware

Software Packs

X-CUBE-AZRTOS-G4 ✓

System Core

Analog

Timers

Connectivity

Multimedia

Security

Computing

Utilities

DMA ✓

ADC1 ✓

TIM1 ✓

I2C2 ✓

CORDIC ✓

GPIO ✓

TIM3 ✓

SPI1 ✓

NVIC ✓

TIM20 ✓

SPI2 ✓

RCC ✓

SPI3 ✓

SYS ✓

USART2 ✓

USART3 ✓

6. Software Pack Report

6.1. Software Pack selected

Vendor	Name	Version	Component
STMicroelectronics	X-CUBE-AZRTOS-G4	2.0.0	Class : RTOS Group : ThreadX SubGroup : Core Version : 6.2.0

7. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32g4_bsd1.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32g4_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32g4_svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-usb-c-pd-solutions-presentation.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32g4-series-product-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf
Brochures	https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32g4.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/fldpstpf11120.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3155-uart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4232-getting-started-with-analog-comparators-for-stm32f3-series-and-stm32g4-series-devices-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4296-use-stm32f3stm32g4-ccm-sram-with-iar-embedded-workbench-keil-mdkarm-stmicroelectronics-stm32cubeide-and-other-gnubased-toolchains-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-

digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5093-getting-started-with-stm32g4-series--hardware-development-boards-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5306-operational-amplifier-opamp-usage-in-stm32g4-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5310-guideline-for-using-analog-features-of-stm32g4-series-versus-stm32f3-series-devices-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5315-stm32cube-firmware-examples-for-stm32g4-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5346-stm32g4-adc-use-tips-and-recommendations-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5738-stm32g4-series-lifetime-estimates-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5325-how-to-use-the-cordic-to-perform-mathematical-functions-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4894-how-to-use-

eeeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signal-to-noise-ratio-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5978-introduction-to-mb1971-llc-hat-12-v-to-75-v1-a-for-f334-g474-nucleo-board-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an2548-introduction-to-dma-controller-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-with-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4908-getting-started-with-uart-automatic-baud-rate-detection-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5156-introduction-to-

security-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5224-introduction-to-dmamux-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5348-introduction-to-fdcan-peripherals-for-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/cd00211314-how-to-optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack2-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4566-how-to-extend-the-dac-performance-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5647-the-goertzel-algorithm-to-compute-individual-terms-of-the-discrete-fourier-transform-dft-in-stm32-products-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an2606-introduction-to-system-memory-boot-mode-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4435-guidelines-for-obtaining-ulcsaiec-607301603351-class-b-certification-in-any-stm32-application-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4657-stm32-inapplication-programming-iap-using-the-usart-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4841-digital-signal-processing-for-stm32-microcontrollers-using-cmsis-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5056-integration-guide-for-the-xcubesbsfu-stm32cube-expansion-package-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5305-digital-filter-

for related Tools & Software	implementation-with-the-fmac-using-stm32cubeg4-mcu-package-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5315-stm32cube-firmware-examples-for-stm32g4-series-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5345-highbrightness-rgb-led-control-using-the-bg474edpow1-discovery-kit-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5360-getting-started-with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5361-getting-started-with-projects-based-on-dualcore-stm32h7-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5418-how-to-build-a-simple-usbp-d-sink-application-with-stm32cubemx-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5426-migrating-graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-550-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5464-position-control-of-a-three-phase-permanent-magnet-motor-using-xcubemcsdk-or-xcubemcsdkful-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dualcore-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5698-adapting-the-xcubestl-functional-safety-package-for-stm32-iec-61508-compliant-to-other-safety-standards-stmicroelectronics.pdf
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5731-stm32cubemx-and-stm32cubeide-threadsafe-solution-stmicroelectronics.pdf

& Software

Application Notes https://www.st.com/resource/en/application_note/an5785-boost-voltage-for-related-Tools-mode-on-bg474edpow1-discovery-kit-stmicroelectronics.pdf

& Software

Application Notes https://www.st.com/resource/en/application_note/an5788-stm32-digital-power-pid-and-iir-filters-for-smmps-control-design-and-comparison-on-bg414edpow1-discovery-kit-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4502-stm32-smbuspmbus-expansion-package-for-stm32cube-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5952-how-to-use-cmake-in-stm32cubeide-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5496-guidelines-for-the-buck-voltage-mode-on-the-bg474edpow1-discovery-kit-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5497-introduction-to-the-buck-current-mode-with-the-bg474edpow1-discovery-kit-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an6179-how-to-integrate-the-stl-firmware-into-a-time-critical-user-application-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an6127-getting-started-with-stm32h7rx7sx-mcus-in-stm32cubeide-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf

Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6382-introduction-to-swd-multidrop-for-stm32-with-stlink-stmicroelectronics.pdf
Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0430-stm32g471xx473xx474xx483xx484xx-device-errata-stmicroelectronics.pdf
Datasheet	https://www.st.com/resource/en/datasheet/dm00528822.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0440-stm32g4-series-advanced-armbased-32bit-mcus-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-microcontrollers-in-bga-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um3167-stm32g4-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf